

Technology Stack

Date	15 July 2026
Team ID	XXXXXX
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3, JavaScript, Tailwind CSS	Provides a responsive, interactive, and user-friendly interface with reusable components.
2	Backend / Server-Side	Python, FastAPI	Handles API requests, business logic, authentication, and communication with AI services efficiently.
3	Database	SQLite, SQLAlchemy ORM	Stores user accounts, loan details, financial records, and AI recommendations securely.
4	AI Integration	Google Gemini API	Generates AI-powered debt analysis, financial insights, and settlement recommendations.
5	Authentication & Security	JWT (JSON Web Tokens), Passlib (bcrypt)	Provides secure user authentication, password hashing, and session management.
6	API Testing	Postman	Used for testing REST APIs and validating backend endpoints during development.
7	Version Control	Git & GitHub	Manages source code versions and enables project collaboration and backup.
8	Deployment	Render	Hosts the FastAPI backend online, making the application accessible over the internet.
9	Development Environment	Visual Studio Code (VS Code)	Used for coding, debugging, and project management with integrated development tools.